

Twist in the Dark

The Cosmic Spin You Can't See



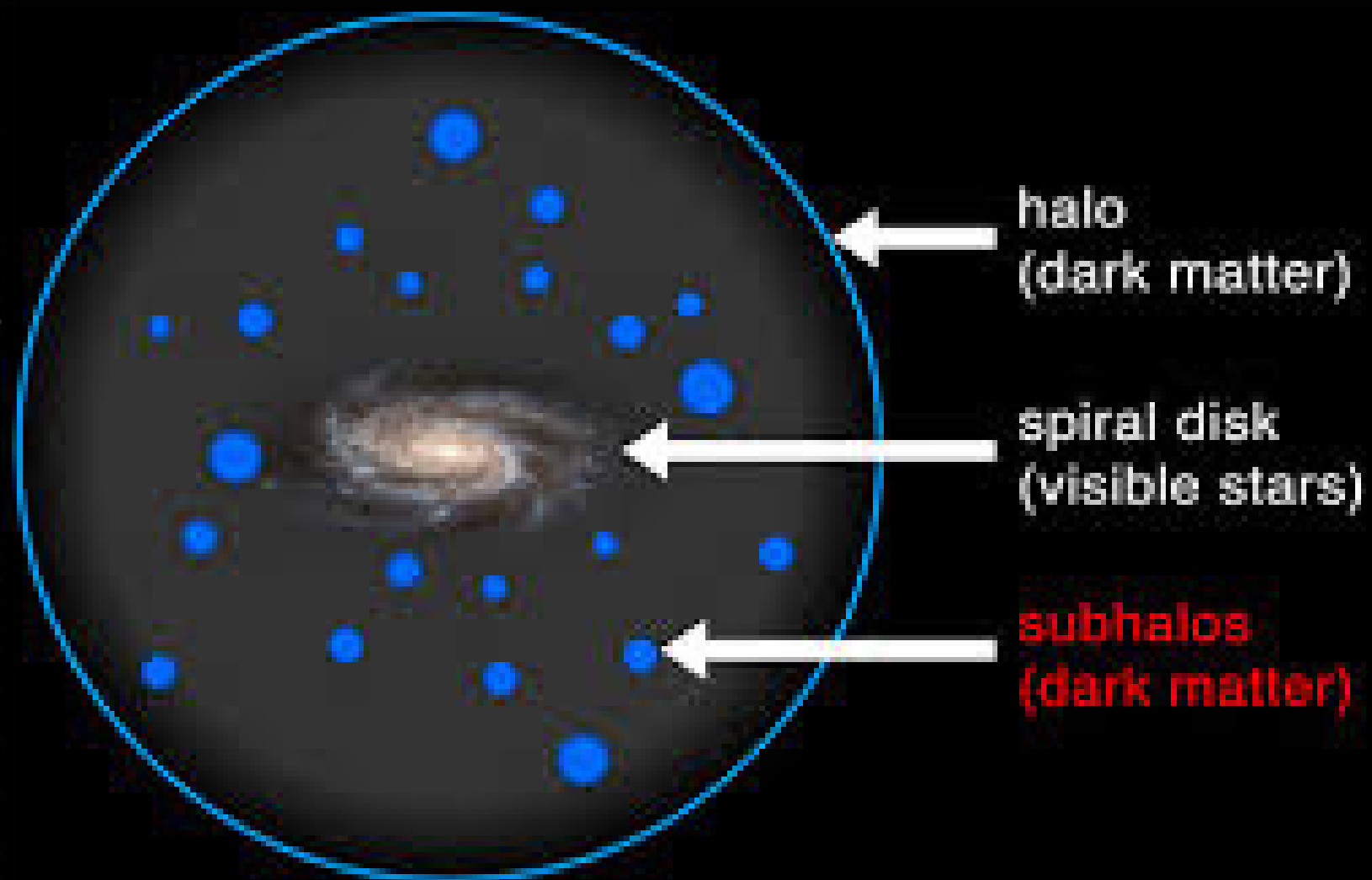
SCAN HERE

WHAT IS DARK MATTER?

- *Dark matter is invisible and makes up most of the universe, unlike ordinary matter that forms stars and planets.*
- *It doesn't emit or reflect light, so we detect it through its gravitational effects on galaxies.*
- *Acting as a hidden framework, it shapes cosmic structures and keeps galaxies held together.*



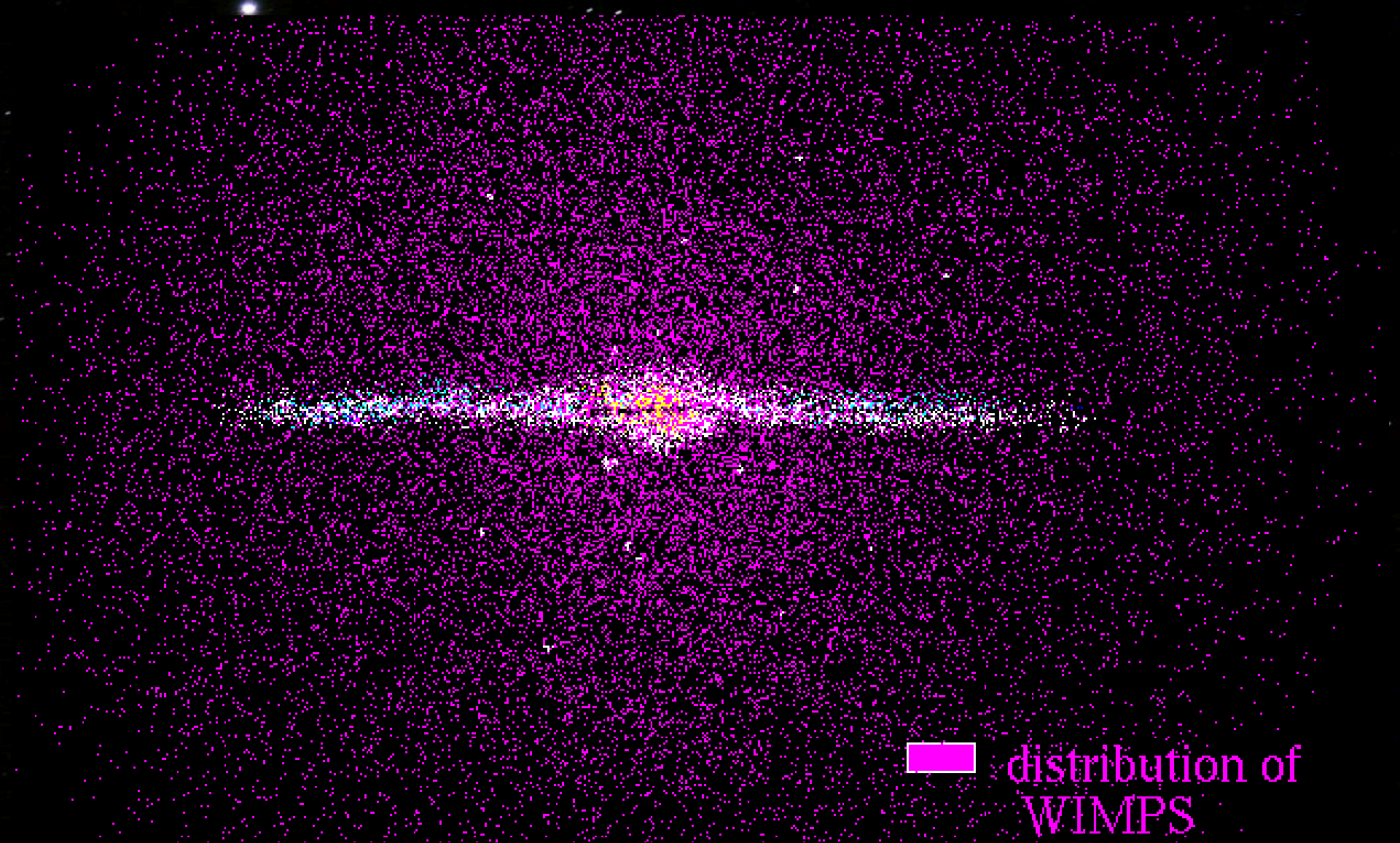
GALACTIC SPIN & DARK MATTER



Galactic spin reveals the presence of dark matter. Stars far from a galaxy's center move faster than expected, defying Newtonian physics. This suggests an invisible halo of mass exerting extra gravity. Dark matter explains why galaxies hold together despite rapid rotation, making rotation curves a key tool to detect it.



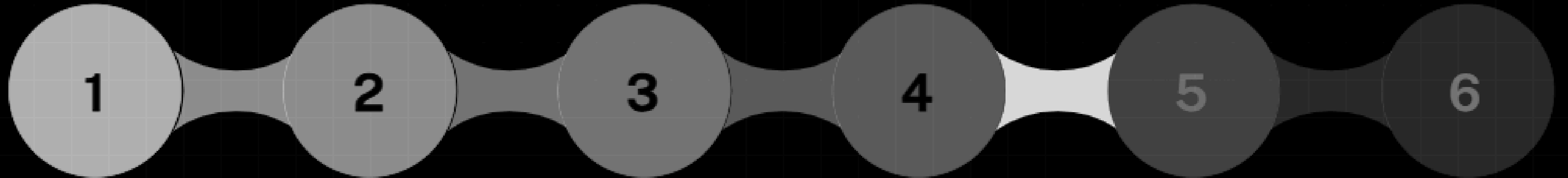
WHAT COULD DARK MATTER BE MADE OF?



(Courtesy: NASA, ESA, A Nierenberg (JPL) and T Treu (UCLA))

DARK MATTER MAY CONSIST OF UNKNOWN PARTICLES BEYOND THE STANDARD MODEL, SUCH AS **WIMPS** (**WEAKLY INTERACTING MASSIVE PARTICLES**), **AXIONS** (**ULTRA-LIGHT PARTICLES**), AND **STERILE NEUTRINOS**. THOUGH INVISIBLE, THESE PARTICLES EXERT GRAVITATIONAL FORCE, HOLDING GALAXIES TOGETHER AND SHAPING THE UNIVERSE, SERVING AS AN UNSEEN BACKBONE OF REALITY.

HOW TO DETECT DARK MATTER



Measure Galactic Spin

Determine the angular momentum of visible matter in the galaxy, such as stars and gas clouds.

Estimate Dark Matter Halo Spin

Use simulations and gravitational effects to infer the spin properties of the surrounding dark matter halo.

Compare the spins

Analyze the alignment and magnitude relationship between the galaxy's spin and its dark matter halo's spin to find deviations.

Detect Anomalies

Look for flat or non-Keplerian rotation curves that indicate additional mass from dark matter affecting galaxy spin.

Assess torque effects

Study the interaction between baryonic matter and dark matter to understand how torques influence halo spin dynamics.

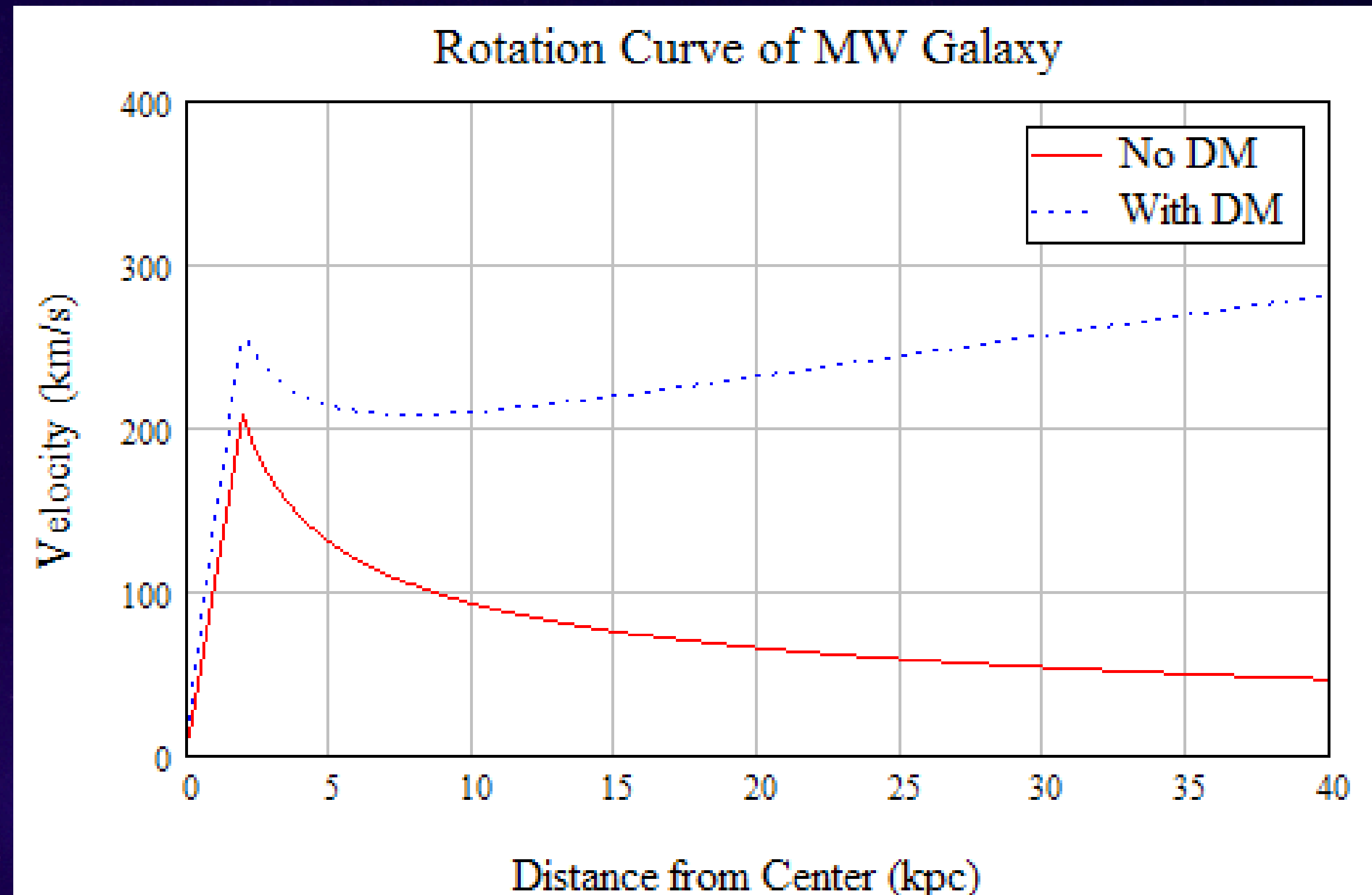
Combine Observations

Integrate spin data with gravitational lensing and cosmological simulations to map and confirm dark matter's presence.

ROTATION CURVES

$$M(< r) = \frac{v(r)^2 r}{G}$$

This equation uses the squared velocity of dark matter gotten from the curve and calculates the mass of dark matter present in that galaxy



SIMULATION

Attached is a simulation of a spiral spinning galaxy with a dark matter halo coded using Python 3.10 and Pygame.



**THIS SIMULATION MIRRORS HOW ASTRONOMERS USE
RADIO DATA AND ROTATION CURVES TO ESTIMATE DARK
MATTER MASS.**

**-THE PURPLE CIRCLE IS THE THEORIZED HALO
SURROUDING
THE GALAXY.**



GET THE SOURCE CODE HERE:

FUTURE APPLICATIONS

NEW PHYSICS, NEW FUTURES

DISCOVERING DARK MATTER COULD UNLOCK HIDDEN LAWS OF NATURE, SPARKING BREAKTHROUGHS IN ENERGY, TECHNOLOGY, AND SPACE EXPLORATION.

INNOVATION THROUGH DATA

THE SEARCH FOR DARK MATTER DRIVES ADVANCES IN AI, BIG DATA, AND SIMULATION — TOOLS THAT POWER THE SCIENCE OF TOMORROW.

A STEP TOWARD UNIFIED SCIENCE

UNRAVELING DARK MATTER COULD BRIDGE QUANTUM PHYSICS AND COSMOLOGY - PAVING WAY FOR A UNIFIED FIELD THEORY.

CREDITS

MOHANNAD KAHIE

Presenter, Premier Academy

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